1.0 COURSEWORK TITLE

Self Defence Academy Management System (SDAMS)

2.0 THE COURSEWORK OVERVIEW

The Self Defence Academy (SDA) is a martial arts institution. It offers disciplines such as Karate, Judo, Taekwondo, MMA and other styles.

Most of SDA's daily operations are managed manually. Information is recorded in different ways. There is no central system. This makes it hard to manage student records, classes, training schedules, payments and facilities.

SDA has appointed your team to develop a Self Defence Academy Management System (SDAMS). You must build it as a JAVA desktop application with a graphical user interface (GUI). The solution must be designed and implemented using object-oriented programming (OOP) principles.

The system supports six user roles. These are Manager, Administrator, Instructor, Student, Facilities Officer, and Attendance Officer. The roles depend on each other through two main workflows:

1. Class workflow. An Instructor creates a class. A Student books the class and pays for it. The Manager sees the payment in the revenue report. The Student may rate the Instructor after the class attendance has been taken by Attendance Officer.
2. Facility workflow. An Instructor reports a facility problem. The Facilities Officer updates the problem until it is resolved.

All features in Section 5.0 are mandatory. They are the minimum expected functionality. Additional features beyond these requirements are encouraged. They earn marks under the Implementation criterion in Section 7.0.

You are required to form a group of six (6) students to design, implement, test and document SDAMS. You are encouraged to review existing martial arts management systems for ideas. Document any logical assumptions you make after your research.

3.0 OBJECTIVES OF THIS COURSEWORK

Develop the practical ability to describe, justify, and implement an Object-oriented system.

4.0 NUMBER OF TEAM MEMBERS

Group Assignment (5 to 6 students)

5.0 SYSTEM FEATURES

Each role must see only its own menu after login.

1. **Managers** should have access to the following functionalities:

Function/Task	Operations	Description
Manage Administrators	Add, view, search, update and delete.	Create, view, search, update and delete administrators.
Staff Report	View and search	Able to view the staff report **(administrator/instructor /facilities officer)** and filter by role and name with total of each role.
Revenue Report	View and search	Able to view and filter the monthly or yearly revenue report, including the number of class registrations and total revenue.

2. **Administrators** should be allowed to view and access the following functionalities:

Function/Task	Operations	Description
Manage Staff	Add, view, search, update and delete.	Create, view, search, update and delete both staff roles

		either instructors or facilities officers.
Manage Students	Add, view, search, update and delete.	Create, view, search, update and delete students.
Payment Records	View and search	Able to view and filter any past payment records made by the student.

3. **Student** should be allowed to view and access the following functionalities:

Function/Task	Operations	Description
Book Classes	Book, view, search, reschedule and cancel.	Book classes made by the instructor. The student also can view, reschedule, and cancel the classes.
Manage Payment	Add, search and view	Make payments for their registered classes. Filter and view their past payment records.
Instructor Rating	Add and view	Able to give and view the scoring rates on the instructor overall classes.

4. **Instructor** should be allowed to view and access the following functionalities:

Function/Task	Operations	Description
Manage Class Schedule	Add, view, search, update and cancel.	Manage training schedules and class availability, allowing students to book available classes conveniently
Issue Facility Problem	Add, view, search, update and delete.	Issue a facility problem such as damaged mats, broken equipment, lighting problems, air-conditioning issues, or damaged rooms.
View Instructor Rating	View and search	View and search the rating score information given by the students.

5. **Facilities officer** should be allowed to view and access the following functionalities:

Function/Task	Operations	Description
Equipment Inventory Management	Add, view, search, update and delete.	Manage the inventory by adding, viewing, searching and updating new and existing martial arts equipment such as mats, punching bags, gloves, uniforms, and other relevant training equipment.
Facility Problem Management	View, search and update.	View, search, and update the facility issues made by the instructor such as damaged mats, broken equipment, lighting problems, air-conditioning issues, or damaged rooms.
Equipment Inventory Stock Reports	View and search	View and search summary information on the equipment stock levels. The stock report can be filtered by equipment information such as the equipment id or name.

6. **Attendance Officer** should be allowed to view and access the following functionalities:

Function/Task	Operations	Description
Manage Class Attendance	Add, view, search and update	View the list of students registered for a scheduled class. Record their attendance as present, absent or late after the class takes place. Update the attendance if a correction is needed.
Attendance Records	View and search	View and search attendance

		records by student, class, instructor, discipline and date
Attendance Report	View and search	Generate and view attendance summaries, including total classes attended, absent and late by student or class

6.0 GENERAL REQUIREMENTS

- The program submitted should compile and be executed without errors
- Validation should be done for each entry from the users to avoid logical errors.
- The implementation code must highlight the use of Object-oriented programming concepts as required by the solution.

7.0 DELIVERABLES:

- Documents delivered softcopy form.
- GUI-driven design is required.
- Submission deadline: **06/12/2026, 11:59PM, Sunday (Week 12)**

7.1 DOCUMENTS: COURSEWORK REPORT

✓ As part of the assessment, you must submit the project report in softcopy form, which should have the following format:

A) Cover Page:
All reports must be prepared with a *front cover*. A protective transparent plastic sheet can be placed in front of the report to protect the front cover. The front cover should be presented with the following details:

- Module
- Coursework Title
- Intake
- Group Number
- Student name and TPnumber

B) Contents:

- Introduction and assumptions.
- Description and justification of Object-oriented programming concepts incorporated into the solution's coding.
- Design solution including Use Case and Class Diagram.
- Screenshots of output of the program with appropriate explanations.

- Screenshots of output of the additional features with appropriate explanations.

C) Limitation, Future Work and Conclusion

D) References

8.0 ASSIGNMENT ASSESSMENT CRITERIA

The assignment assessment consists of three components: Program Solution & Design (60%), Documentation (20%) and Presentation (20%). Details of the division for each component are as follows:

<table>
<tr><th colspan="2">Program Solution and Design (60%)</th><th colspan="2">Documentation (20%)</th></tr>
<tr><th>Criteria</th><th>Marks Allocated</th><th>Criteria</th><th>Marks Allocated</th></tr>
<tr><td>Program Solution : Performance result of all operations and design as per requirements</td><td>40%</td><td rowspan="2">Structure of the report and references
Content:
• Usecase and Class Diagram with Descriptions
• Description and justification of the object-oriented concepts incorporate
• Program output screenshots
• Project description, limitation and conclusion</td><td rowspan="2">20%</td></tr>
<tr><td>Program Design: Appropriate design (Usecase & Class Diagrams) and the implementation of codes illustrating the object oriented programming concepts incorporated</td><td>20%</td></tr>
<tr><th colspan="4">Presentation (20%)</th></tr>
<tr><th colspan="3">Criteria</th><th>Marks Allocated</th></tr>
<tr><td colspan="3">• Ability to answer questions addressed by the lecturer pertaining to the work done and presented</td><td>20%</td></tr>
</table>

9.0 DEVELOPMENT TOOLS

The program developed for this assignment must be implemented using **Object-Oriented Programming (OOP) principles and the Java programming language only**. A **Graphical User Interface (GUI) must be developed using Java Swing only**.

Text files must be used as the data storage mechanism for the system. All required system data must be stored, retrieved, updated and managed using text files. The use of other programming languages or database management systems is not permitted.

10.0 ACADEMIC INTEGRITY

- You are expected to maintain the utmost level of academic integrity during the duration of the course.
- Plagiarism is a serious offence and will be dealt with according to APU and regulations on plagiarism.

11.0 AI USAGE POLICY FOR CODING ASSIGNMENTS

This is an Amber assessment. You may use AI tools (e.g., ChatGPT, Copilot, Gemini, etc.) only for the following supportive purposes:

- Understanding programming concepts, syntax, or JAVA language features.
- Identifying relevant keywords or technical terms.
- Clarifying compiler error messages or documentation terminology
- Debugging assistance, locating and understanding errors in code you have written yourself
- Language refinement for code comments, documentation, or report write-ups

AI must not be used for:

- Generating complete or partial source code, algorithms, or program logic
- Producing final solutions to any of the assignment tasks
- Writing entire functions, classes, or data structure implementations on your behalf

You must ensure that:

- All submitted code is your own original work and reflects your own understanding
- All AI-assisted content is verified for accuracy
- You can explain and justify every part of your code during evaluation or questioning

Declaration & Integrity:

- If AI tools are used, you must clearly declare the purpose of usage. For example: "AI was used to identify appropriate keywords and concepts related to file handling in JAVA programming"
- Copying, paraphrasing, or adapting AI-generated code, whether modified or unmodified, is considered a violation of academic integrity.
- Lecturers reserve the right to conduct oral questioning, code tracing, or live demonstrations to verify authenticity.